Dengue Fever Forecasting

Thailand · 2009–2025

A time-series forecasting study to support pharmaceutical supply chain decisions in the dengue endemic region of Southeast Asia

Group 8

Jaganatha Pandiyan (25BM6JP18)

Nemani Siva Sankar (25BM6JP31)

Solanki Preya Girishchandra (25BM6JP50)

Sumit Saurabh (25BM6JP55)

Vinamra Choudhary (25BM6JP58)

What We'll Cover

01 **Why This Project?**

The business case for dengue forecasting in pharma supply chains

02 **Understanding the Data**

16 years of Thailand dengue data and what the numbers tell us

03 **Decomposing the Signal**

Separating trend, seasonality, and noise

04 **Building Baseline Models**

Starting simple with AR and MA benchmarks

05 **The SARIMA Framework**

Model selection, parameter search, and feature engineering

06 **Results & Business Implications**

What the model delivers and what it can't

Why Dengue?

The business case for forecasting in an endemic disease market

400M+

dengue infections worldwide every year (WHO estimate)

$8.9B

global economic burden annually, mostly in Asia-Pacific

6–8 wks

lead time pharma companies need before an outbreak peak to prepare inventory

The Supply Chain Problem

Dengue is not a steady-state disease. It explodes in monsoon months (May through September) and collapses the rest of the year. Pharmaceutical distributors and hospital procurement teams face a cruel dilemma every year:

- Order too early → capital tied up in unused stock
- Order too late → shortages during peak demand
- Guess wrong → patient outcomes and revenue both suffer

Why Thailand?

A case study in a high-burden, data-rich dengue market

High Incidence

Thailand averages **60,000 – 100,000** cases per year, creating meaningful commercial demand for treatments and diagnostics

High Cost

The cost burden was estimated at **$440 million** a year, and **~7 lost workdays** for those infected *(Thisyakorn et al)*

Recent Disruption Event

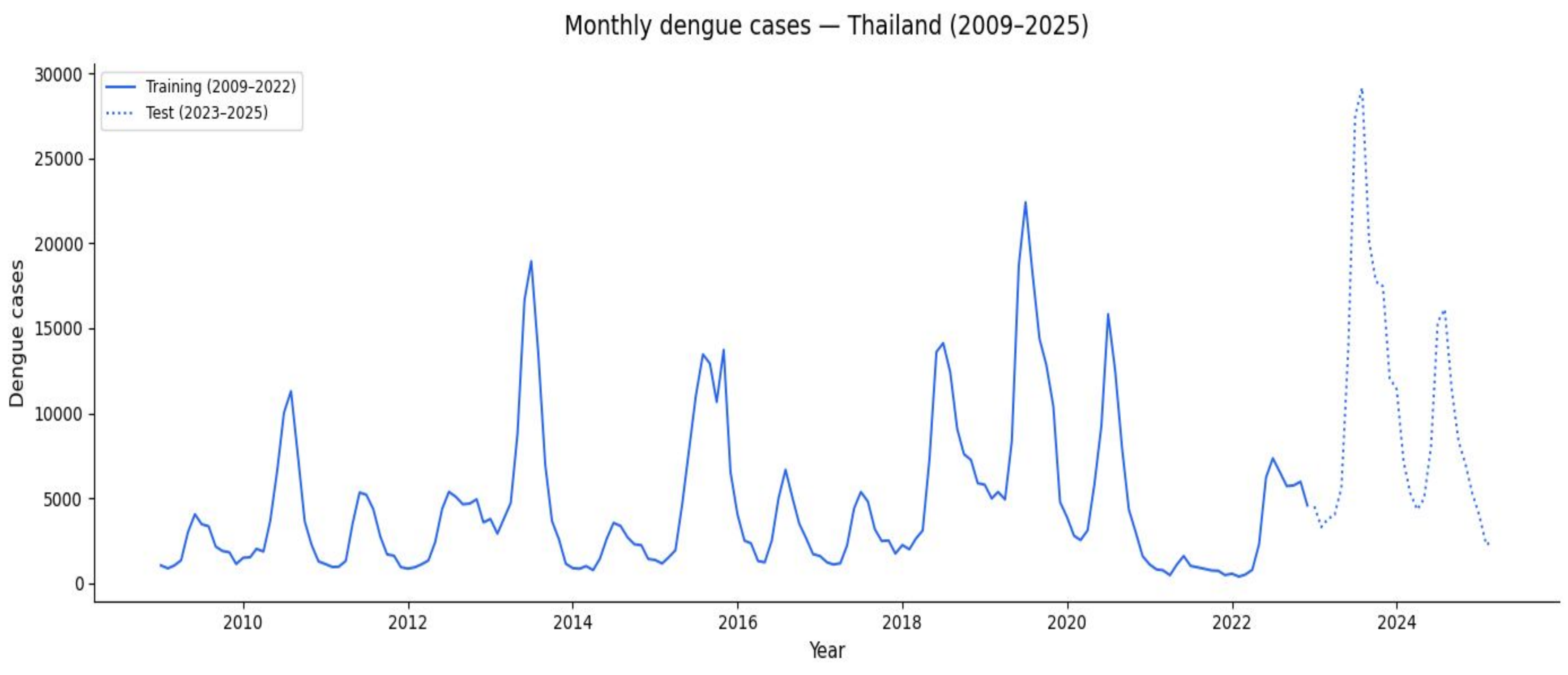
The 2023 super-outbreak (29,000 cases in a single month) provides an extreme stress test for any forecasting model.

Consistent Reporting

Thailand has reported monthly dengue surveillance data publicly since 2009

16 Years of Monthly Cases

OpenDengue dataset — Thailand (Jan 2009 to early 2025)



Annual spikes

Every year shows a distinct monsoon-driven peak — the seasonal signal is unmistakable.

2021–2022 dip

COVID-19 lockdowns suppressed reported cases — a statistical artefact we address later.

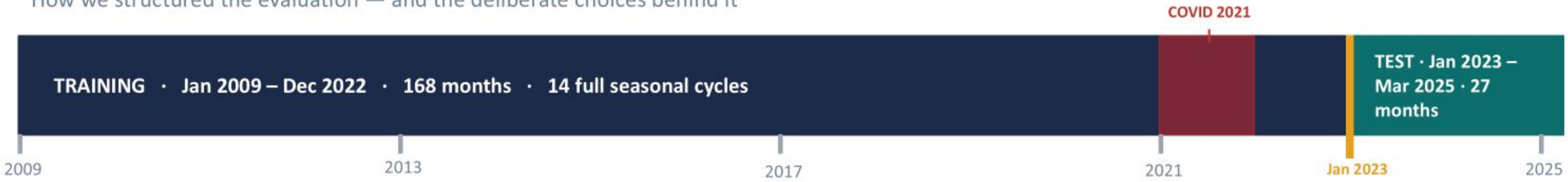
2023 super-peak

~29,000 cases in a single month — the largest on record. The ultimate stress test.

Dataset: 195 monthly observations · No gaps · Source: OpenDengue / Nature Scientific Data

Train / Test Split

How we structured the evaluation — and the deliberate choices behind it



Why not include 2023 in training?

2023 is the hardest forecasting challenge in the dataset — a once-in-16-year super-outbreak peaking at 29,000 cases in a single month. Including it in training would let the model "see the answer" and inflate accuracy scores.

Keeping it in the test set means every MAPE number is earned on genuinely unseen extreme data — the most honest evaluation possible.

Why not include 2022 in the test set?

Since we imputed 2021 values by replacing COVID-suppressed cases with historical monthly averages, the 2022 data — which sits immediately after this imputed period — is structurally contaminated by that intervention. Evaluating the model on data that follows an artificially reconstructed year would produce unreliable test metrics.

To ensure a clean, uncontaminated evaluation, the test set begins at Jan 2023 — the first full year after both the COVID disruption and its imputation window have passed.

Why 27 months in the test set?

27 months covers two complete monsoon peaks — July 2023 and July 2024 — plus the off-season troughs between them. A single-season test could be lucky or unlucky.

Two full seasonal cycles produce a variance-stable MAPE estimate. The number you see in results is not a one-season fluke — it reflects the model's true consistency across different outbreak magnitudes.

168 training months

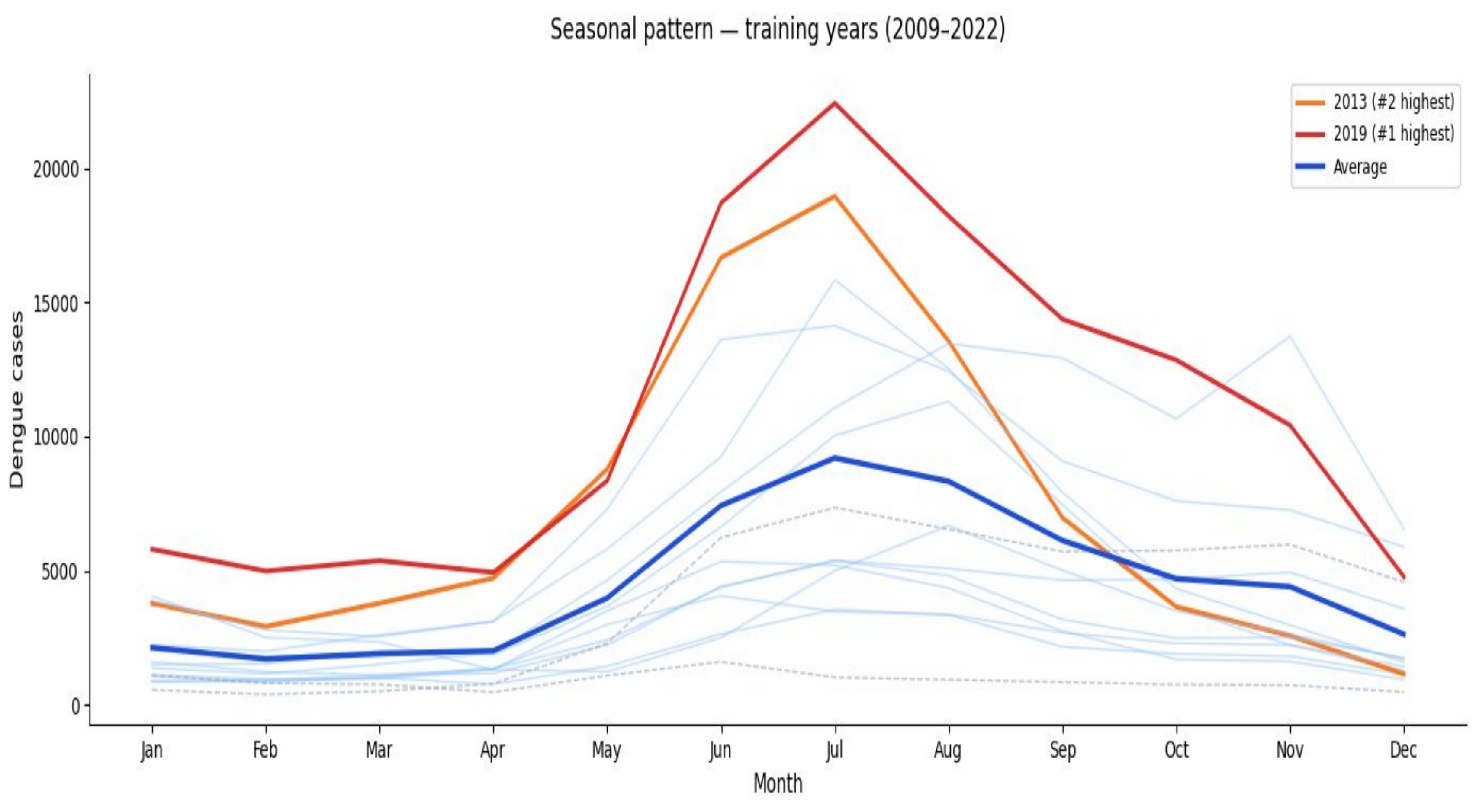
14 full seasonal cycles in train

27 test months

2 complete outbreak peaks tested

Every Year, the Same Story

Seasonal overlay — each line represents one calendar year

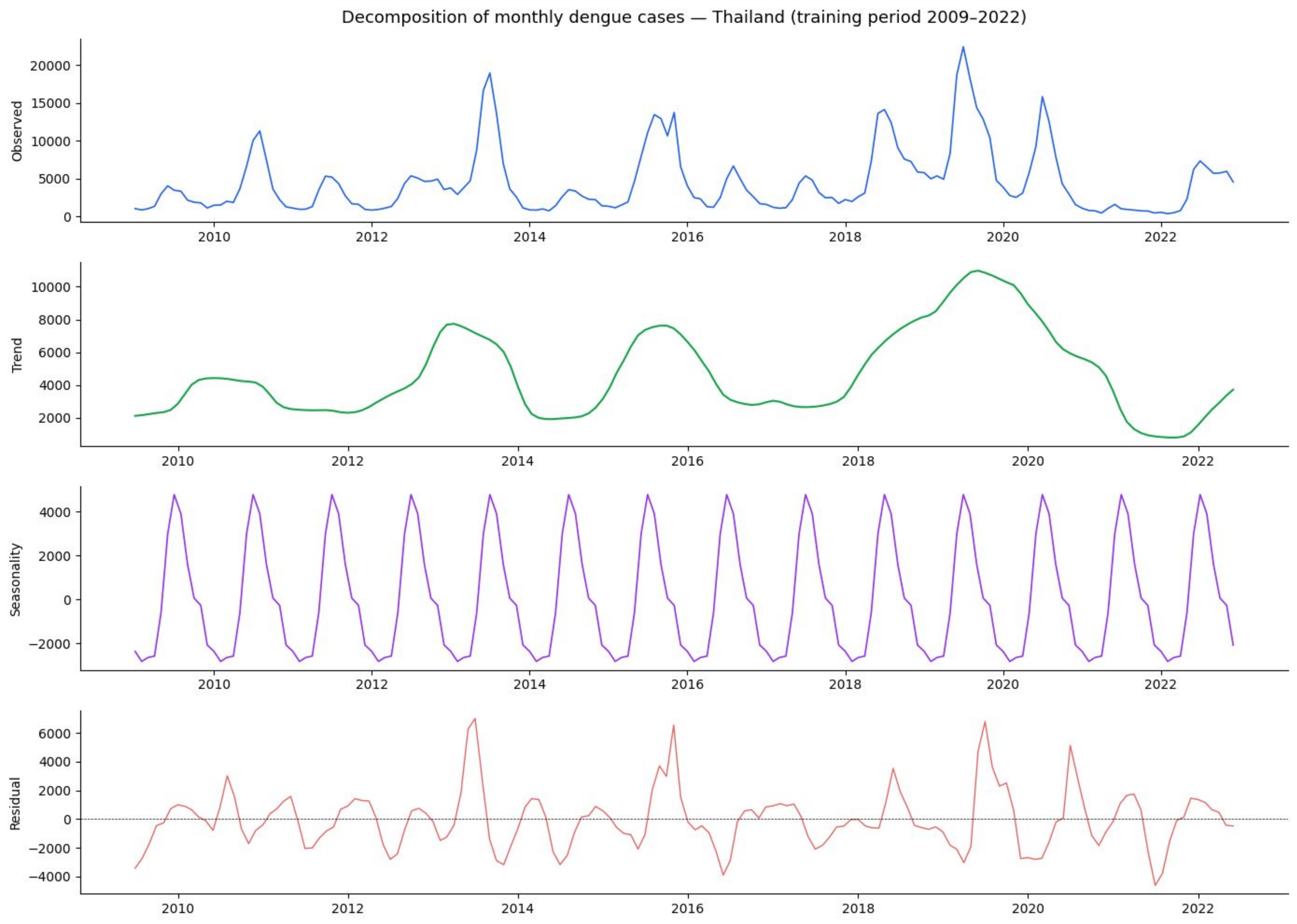


What this tells us

1. **Peak:** July – August every single year without exception
2. **Trough:** March – April, cases fall close to zero
3. 2019 (red) and 2013 (orange) were ~2 - 2.5× above average
4. COVID years (dashed) are clear outliers, not a structural shift
5. The average (blue) shows how reliable the seasonal shape is

Decomposing the Signal

Additive decomposition separates the three forces driving dengue case counts



Trend

Slow multi-year cycles — no permanent upward direction

Seasonality

Clear annual wave peaking in July – Aug every year

Residual

Random shocks. The 2021 anomaly is clearly visible here

Is the Series Stationary?

We run two tests with opposite null hypotheses - agreement between both eliminates ambiguity

ADF Test (Augmented Dickey-Fuller)

H_0: $\gamma = 0$ (Unit root present / Non-stationary)
H_a: $\gamma < 0$ (Stationary)

ADF-stat = $\hat{\gamma} / SE(\hat{\gamma})$

Test Statistic (ADF)	p-value	Verdict
−7.0292	< 0.0001	STATIONARY ✓

ADF - Mathematical Terms & Statistic

- **$\Delta y_{\square}$:** First difference of the series at time t.
- **γ:** The focus of the test; if γ=0, a unit root exists.
- **DF-stat:** $\hat{\gamma} / SE(\hat{\gamma})$. Test statistic follows Dickey-Fuller distribution.
- **$\varepsilon_{\square}$:** White noise error term.

Formula: $\Delta y_{\square} = \alpha + \beta t + \gamma y_{\square-1} + \sum \delta_{\square} \Delta y_{\square-\square} + \varepsilon_{\square}$

KPSS Test (Kwiatkowski-Phillips-Schmidt-Shin)

H_0: $\sigma^2\varepsilon = 0$ (Stationary / No unit root)
H_a: $\sigma^2\varepsilon > 0$ (Non-stationary)

$LM = \sum S_{\square}^2 / (T^2 \cdot s^2(l))$, where $S_{\square} = \sum \varepsilon_i$

Test Statistic (KPSS)	p-value	Verdict
0.1347	0.10	STATIONARY ✓

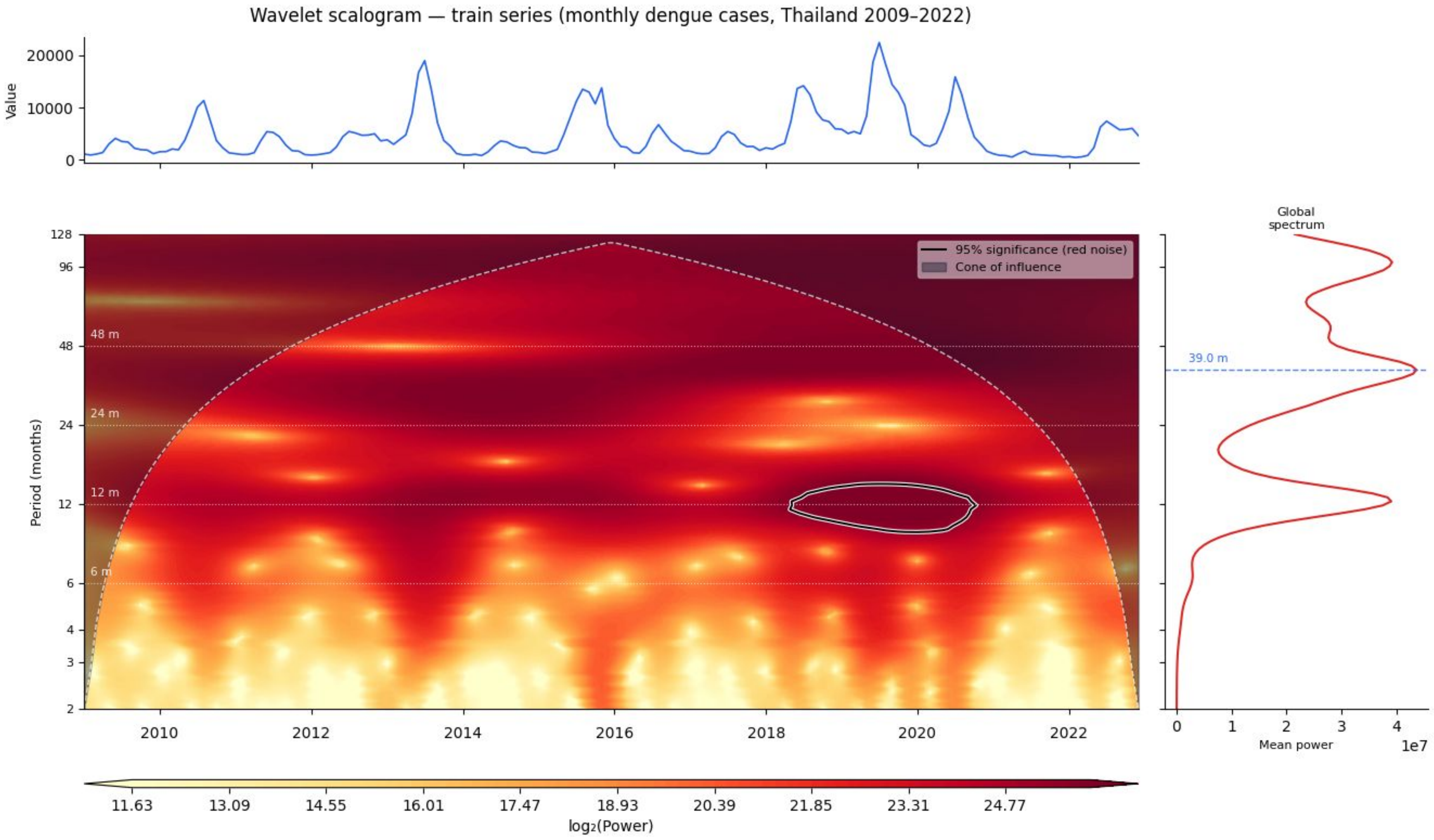
KPSS - Mathematical Terms & Statistic

- **LM:** Lagrange Multiplier test statistic.
- **$S_{\square}$:** Partial sum of residuals ($S_{\square} = \sum \varepsilon_i$).
- **$s^2(l)$:** Estimator of long-run variance.
- **ε_i:** Residuals from regression.
- r_t = random walk component

Formula: y_t = ξt + r_t + ε_t

What this means for modelling: Both tests confirm stationarity → d = 0, no regular differencing needed. SARIMA can model the series as-is without distorting the data through differencing.

Seasonality Patterns (1/2)



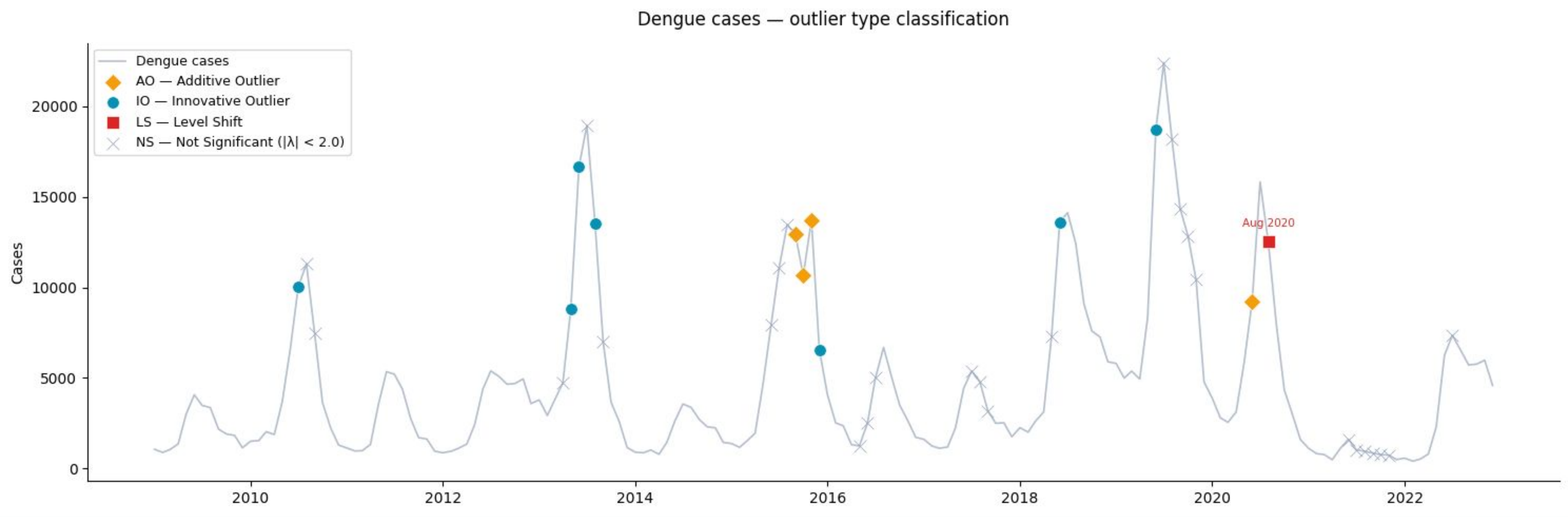
Statistically significant **annual seasonality** is observed, indicating a recurring yearly dengue transmission

Seasonal intensity varies across time, with the strength of the annual cycle changing between years, suggesting temporal instability in outbreak magnitude.

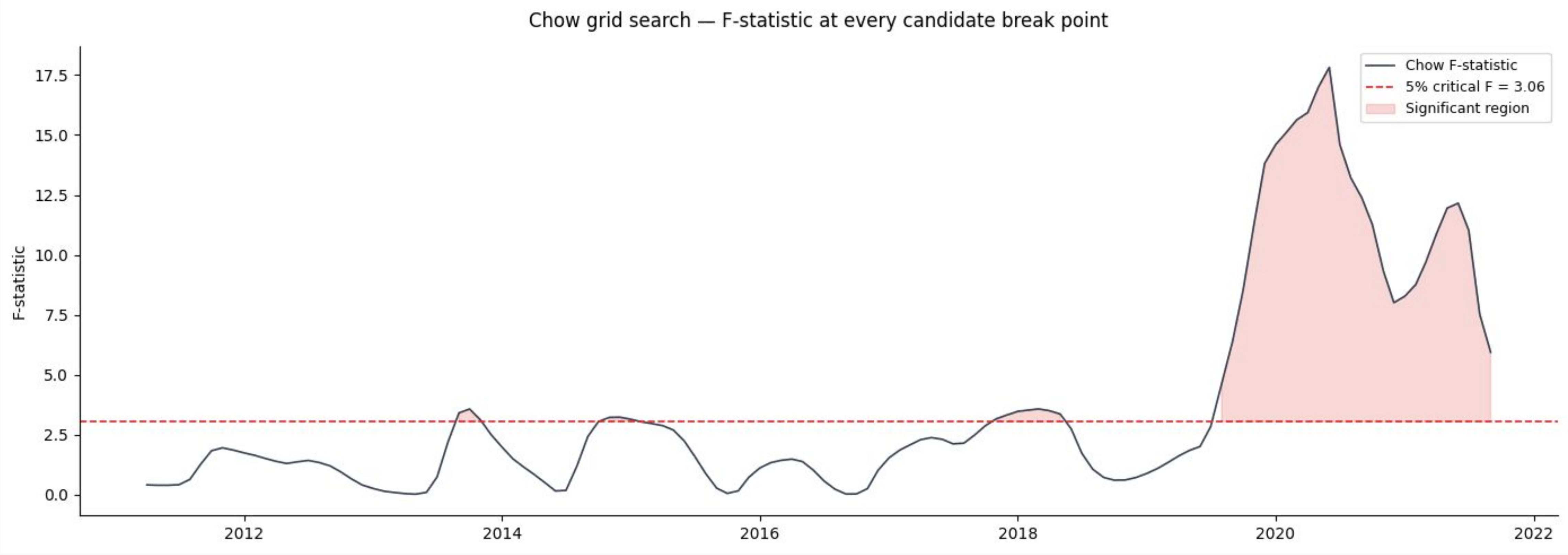
Seasonality Patterns (2/2)

```
==========================================================================================
  HEGY COMPARISON — As is vs Imputed
==========================================================================================
  Frequency / Root        Period           p-val (as is)  Result (as is)           p-val (imputed)  Result (imputed)
------------------------------------------------------------------------------------------
  Zero (Long-run)         Non-seasonal           0.0111  Stationary                        0.0407  Stationary
  k = 1 (Annual)          12 months              0.0150  Stationary                        0.1936  Unit Root (D=1)
  k = 2 (Semi-Ann.)       6 months               0.0007  Stationary                        0.0167  Stationary
  k = 3 (Tri-Ann.)        4 months               0.0009  Stationary                        0.1047  Unit Root (D=1)
  k = 4 (Quarterly)       3 months               0.0000  Stationary                        0.0000  Stationary
  k = 5 (Bi-Mon.)         2.4 months             0.0000  Stationary                        0.0000  Stationary
  k = 6 (Nyquist)         2 months               0.0001  Stationary                        0.0003  Stationary
```

Are there Outliers in the Series?



Are there Breaks in the Series?



Handling the COVID Anomaly

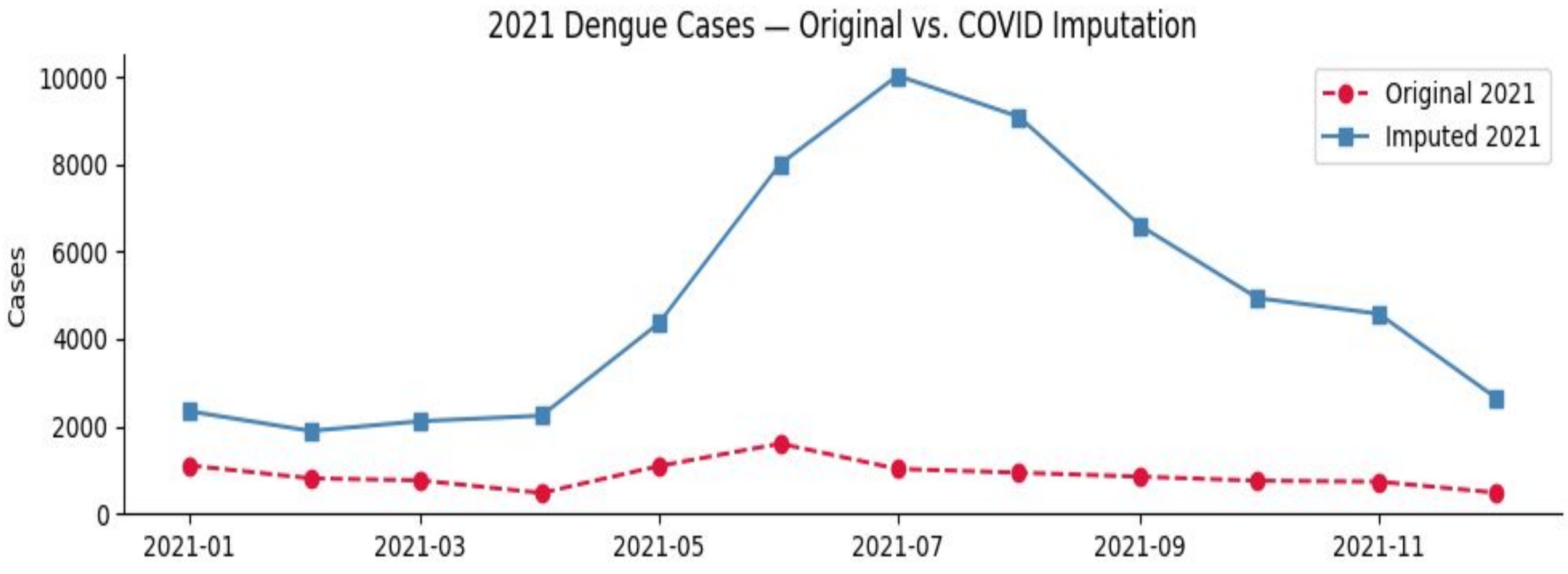
The Problem

Thailand's COVID-19 lockdowns in 2021 suppressed hospital visits and reporting, making dengue cases appear near zero for the entire year.

If our model learns that 2021-style low-case periods are "normal", it could be biased toward underestimating future outbreaks.

The Fix: Historical Mean Imputation

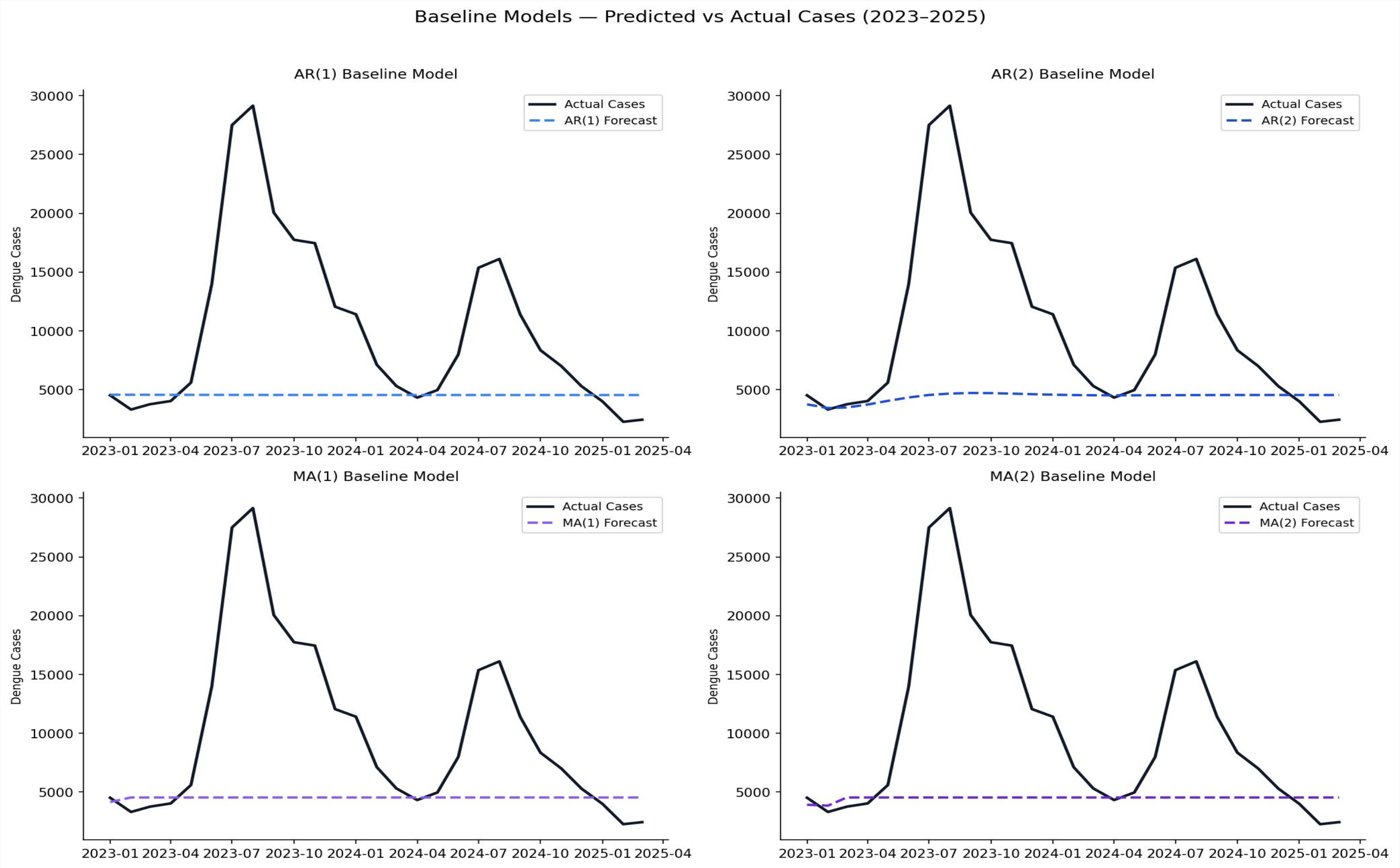
For each month in 2021, we replace the suppressed value with the average of the same calendar month across 2009 – 2020. This restores the expected seasonal pattern without introducing future data leakage.



Imputed 2021 now shows ~10,000 peak cases in July, consistent with the 2009–2020 seasonal average.

Starting with Baselines

Before building complex models, we need to know how simple ones perform as they set the floor



AR(1), AR(2), MA(1), MA(2)

All four models forecast a near-flat line around 4,500 cases per month, completely missing the 2023 outbreak that reached 29,000.

Why they fail

These models look at recent values and recent errors, and they have no concept of seasonality. They anchor around the mean and cannot anticipate the monsoon surge.

Typical RMSE: ~9000

We need seasonal models to model the series better

Train: 2009–2022 · Test: 2023–2025

Building the SARIMA Models

Grid search + lag feature engineering to find the best specification

Step 1: Parameter Search

Rather than guessing parameters from ACF/ PACF plots, we run an exhaustive AIC-ranked grid search over all combinations of $p \in \{0, 1, 2, 3\}, q \in \{0, 1, 2, 3\}, P \in \{0, 1, 2\}, Q \in \{0, 1, 2\}$.

Winner: $\boldsymbol{SARIMA(1, 0, 1)(1, 0, 1, 12)}$

Step 2: Lag Feature Screening

Baseline SARIMA achieved ~44% MAPE. The model knew the seasonal shape but not the current momentum. It couldn't detect whether an outbreak was already building.

We tested lags 1, 2, 3, 4, 6, 8, and 12 as exogenous features. Only lag_1 (last month's case count) produced a meaningful improvement

Lag	RMSE	Verdict
None	7890.87	Baseline
lag_1	3458.55	−4432
lag_2	6438.34	−1452
lag_3	8457.82	+567
lag_12	8983.29	+1092

Key insight

Adding lag_1 dropped RMSE from 7890.87 to under 3500, a 44 percentage point reduction. This single feature captures outbreak momentum: when cases are rising fast this month, next month will likely be high too.

The 5-Model Shootout

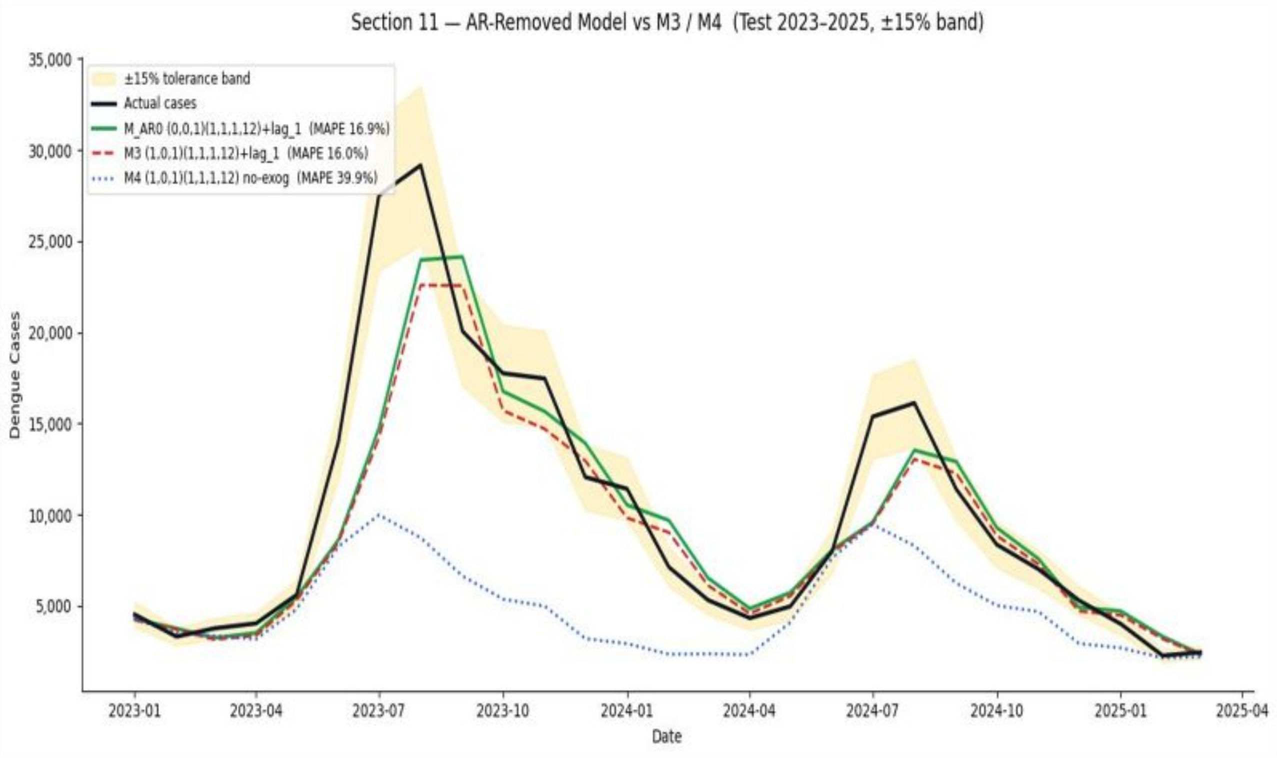
Five SARIMA variants tested

Model	Configuration	Key Feature	Train RMSE	Train MASE
M1	$SARIMA(1,0,1)(1,0,1,12)$	+ lag_1	1570.11	0.254
M2	$SARIMA(0,0,1)(1,1,1,12)$	+ lag_1	1503.32	0.231
M3	$SARIMA(1,0,1)(1,1,1,12)$	+ lag_1 (D=1)	1579.25	0.256
M4	$SARIMA(1,0,1)(1,1,1,12)$	No exog	1648.58	0.281
M5	$SARIMA(1,0,1)(1,0,1,12)$	Log + lag_1	12462729.88	281.015

The D=1 insight: Seasonal differencing (D=1) was the single most impactful structural choice. M3 with D=1 outperformed its M1 counterpart despite identical p,q,P,Q orders. Models with seasonal differencing consistently beat those without it

How Well Do The Two Best Models Track?

Both winning models tested on held-out 2023–2025 data



$SARIMA(0,0,1)(1,1,1,12) + lag\_1$

3469 Test RMSE

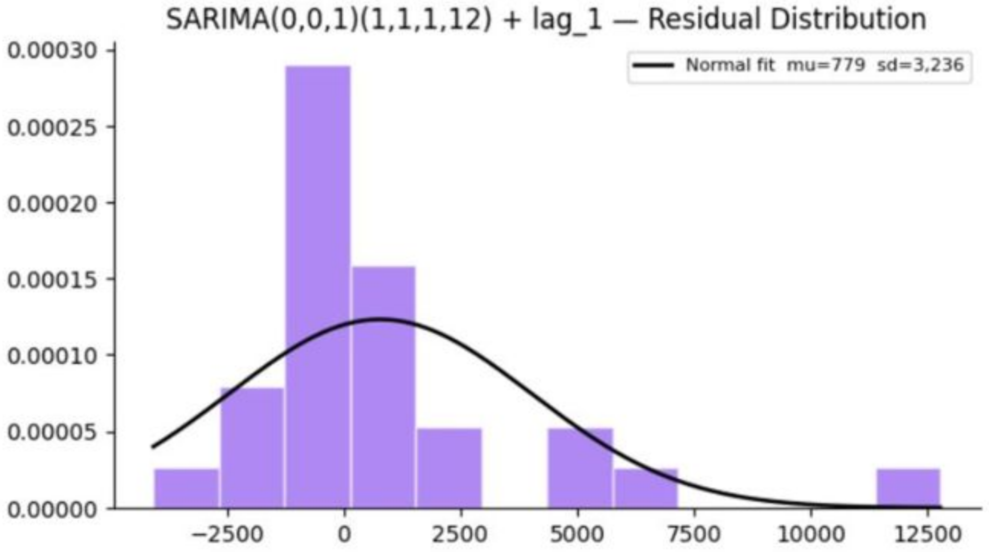
$SARIMA(1,0,1)(1,1,1,12) + lag\_1$

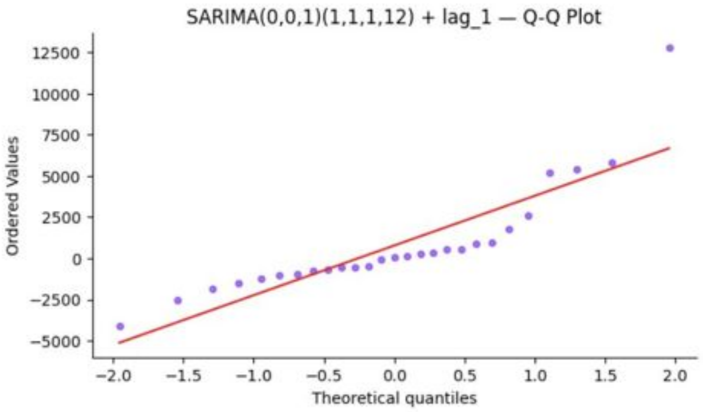
3571 Test RMSE

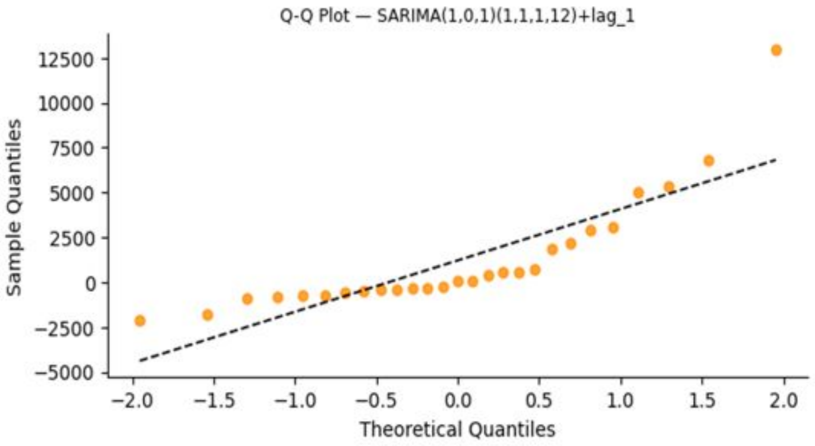
Residuals Analysis

Is there anything we missed in our modelling?

$Metric$	$SARIMA(0,0,1)(1,1,1,12)\_LAG\_1$	$SARIMA(1,0,1)(1,1,1,12)\_LAG\_1$
Shapiro-Wilk	0.0001 (p-value)	0.0000 (p-value)
Ljung-Box	0.5670 (p-value)	0.5519 (p-value)
Mean of residuals	779.1	1217
Std of residuals	3236	3147
Skewness	1.9	2.14
Kurtosis	4.70	4.68

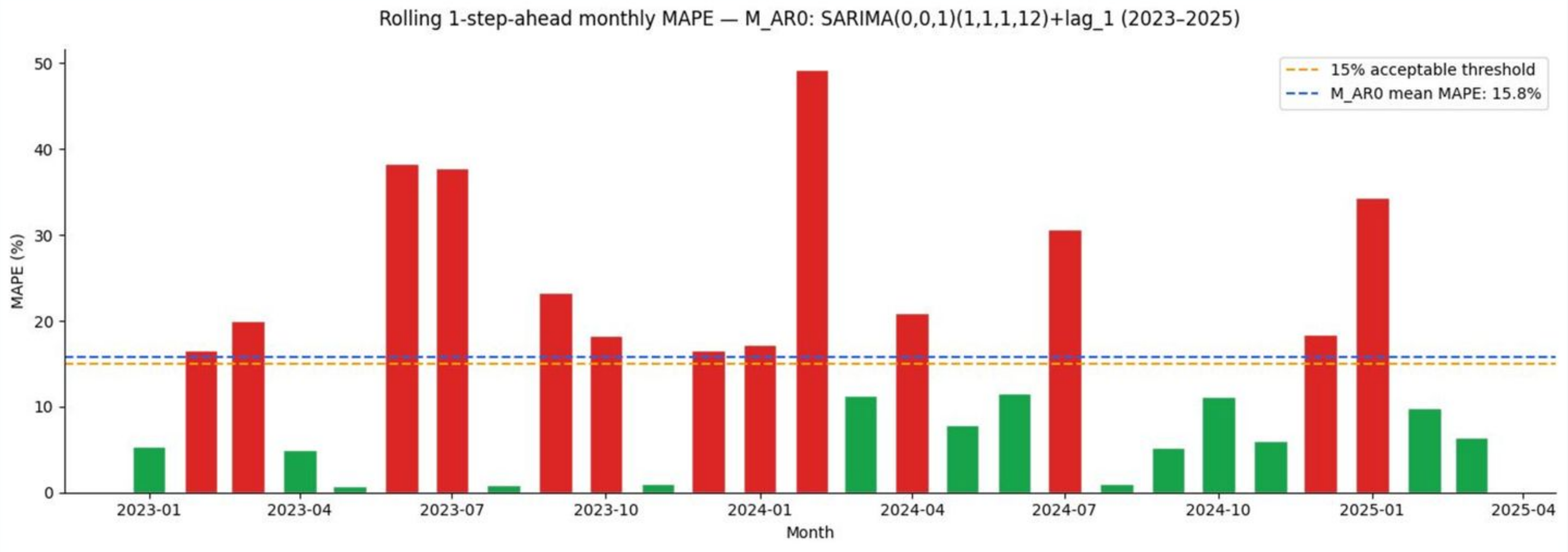






Month-by-Month Accuracy

Not all months are equal — the model performs very differently during outbreak vs. quiet periods



2023		2024		2025	
RMSE	**6108**	RMSE	**2350**	RMSE	**361**
MASE	1.213	MASE	0.474	MASE	0.066

Final Model Performance

Model	Test RMSE	Test MASE
Baseline Models $AR(1), AR(2), MA(1), MA(2)$	~9000	~1.62
$\boldsymbol{SARIMA(0,0,1)(1,1,1,12) + lag\_1}$	3469	0.514
$\boldsymbol{SARIMA(1,0,1)(1,1,1,12) + lag\_1}$	3571	0.832

1.62 → 0.514

MASE improvement from baselines to final model

Driven by two decisions:

1. Adding lag_1 (last month's cases) as an exogenous momentum signal
2. Seasonal differencing D=1 to correctly model the annual cycle

What This Means for the Business

Translating model accuracy into operational decisions for pharma supply chains

Inventory Planning

A 3469 RMSE means the model's forecast is within roughly 2000 - 3000 cases. For procurement, that's a reliable enough signal to order 6–8 weeks ahead of the peak.

Seasonal Readiness

The model consistently identifies the July–August peak window. Even if the exact magnitude is uncertain, supply chain teams can trigger inventory buildup in May every year with high confidence.

Outbreak Early Warning

Because lag_1 captures momentum, a sudden spike in March or April — like 2023 — registers immediately in the forecast for April/May. The model acts as an early warning system.

Budget Forecasting

Annual dengue burden estimates for budget allocation can now be grounded in statistical forecasts rather than rule-of-thumb assumptions or last year's numbers.

Honest Limitations

Where the model struggles — and why that's expected

Extreme peaks

The 2023 super-outbreak (~29,000 cases) is nearly 3× the historical average. No univariate model trained on past data can fully anticipate a black swan event. The model correctly captured the direction and timing — but not the full magnitude.

Univariate only

This model uses only past case counts. Adding climate data (rainfall, temperature), vector control programme data, or population mobility signals could materially improve peak-month accuracy.

COVID structural break

2020–2022 are genuinely anomalous years. While we handled 2021 through imputation, the post-COVID re-emergence dynamics may have shifted underlying disease transmission patterns in ways the model hasn't fully captured.